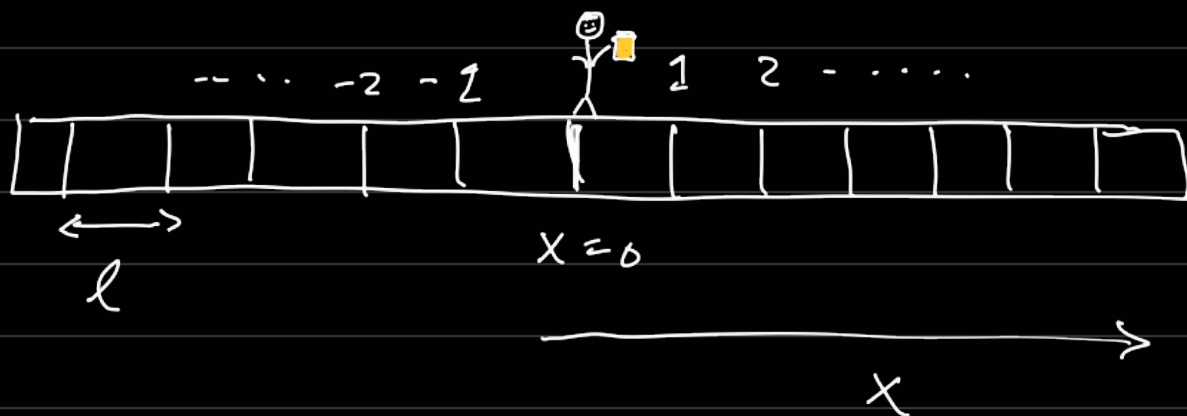


\* Go over some basics about probability and understand where everything comes from.  
→ Best way to do this is with the concept of a random walk:



- Called a random walk as we imagine a drunk starting at  $x=0$  and then stumbling about from left-to-right (randomly) with equal length for each step.  
This system actually represents many basic physical systems such as
  - i) Magnetic Spin
  - ii) Diffusion of gas
  - iii) light intensity
  - etc....
- We could look at this system in detail but I'll only cover enough so the concepts can be easily transferred.

Idea: He will take some number of steps to the right ( $n_R$ ) and some number of steps to the left ( $n_L$ ) so a total number of steps  $N$

$N$ : Total number of steps

$n_R$ : Number of steps to the right  
 $n_L$ : Number of steps to the left

$$N = n_R + n_L$$

- Now,  $N$  does not tell us where the person will be after  $N$  steps. Instead, the relative distance is given by

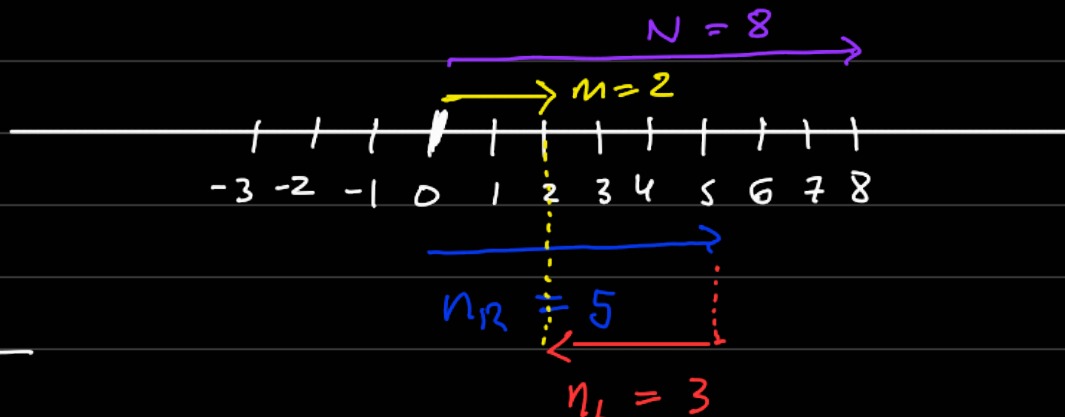
$$m = n_R - n_L \quad \left( \text{Also called the Displacement} \right)$$

example: Suppose we saw the person take  
 $n_R = 5$  (5 steps to the right)  
 $n_L = 3$  (3 steps to the left)

then

$$N = 5 + 3 = 8 \quad (\text{total steps})$$

$$m = 5 - 3 = 2 \quad (\text{relative distance})$$



\* Now note there are assumptions here:

- (i) Note that we did not have to know the order at which the drunk made it to  $m=2$ . Perhaps they could have went  $R L R L R L R R$   
 $\quad \quad \quad \uparrow \text{Right} \quad \uparrow \text{Left}$   
or  $L L L R R R R R$  etc....

This assumption is called statistical independence

Meaning each step is independent of the last.

- When we have such a system we can say each step is characterized by the following probabilities:

$P \equiv$  prob. that a step is to the RIGHT

$q = (1-p) \equiv$  prob that a step is to the LEFT

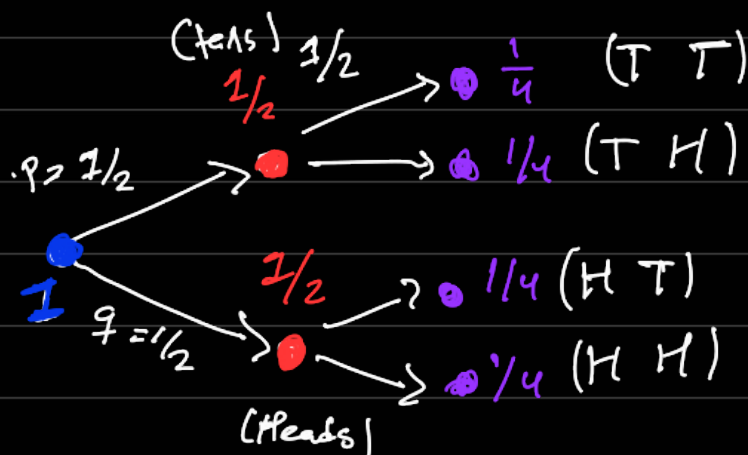
Since probabilities sum to 1 we have

$$p + q = 1$$

(ii) Statistically independent probabilities are multiplied:

To see this we look at a tree diagram

- Let's start with  $p = 1/2 = q$ .



- 1 | Let's suppose we're flipping a coin which has  $p = \text{"heads"} = 1/2$   
 $q = \text{"tails"} = 1/2$
- is the coin before we flip it.
- 1/2 | Now, if we toss the coin it lands on either heads or tails  
 $1/2$   $1/2$

Note that the sum of these two must still be 1, so

$$\frac{1}{2} + \frac{1}{2} = 1$$

- $\frac{1}{4}$  Now, if we toss it yet again we have 4 different outcomes. We can only have 1 of them, giving us a prob. of  $\frac{1}{4}$

Note again that these add to

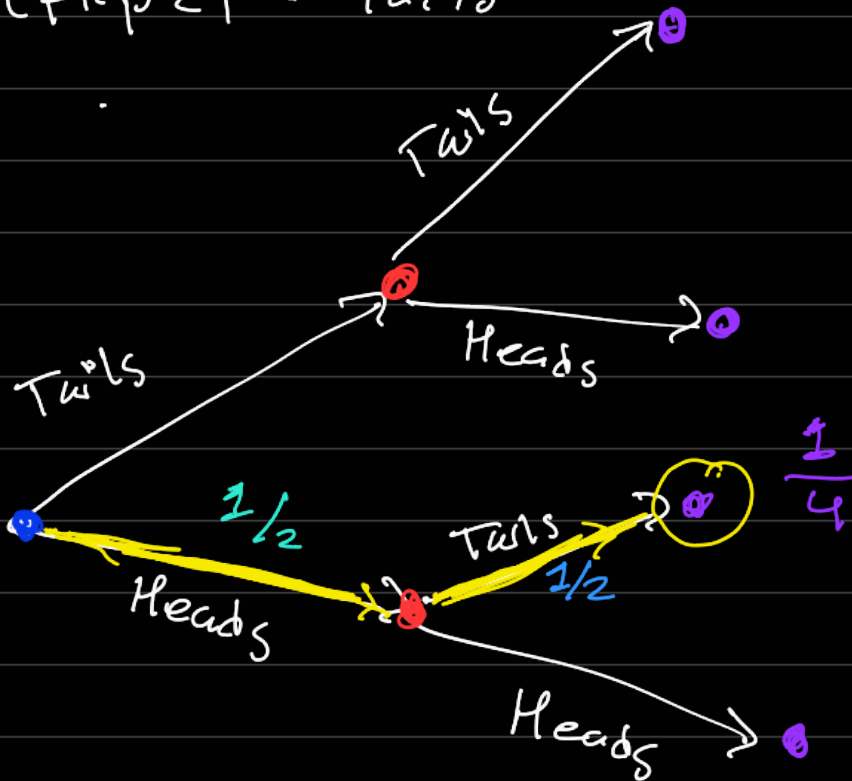
$$1 = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$$

TT TH HT HH

\* Soooo how does this show that Successive tosses have probabilities which multiply?

Well, suppose that we had the result:

- { [flip 1]: Heads
- { [flip 2]: Tails



→ This means we follow the path along           

- Each step has probability of  $\frac{1}{2}$ , so

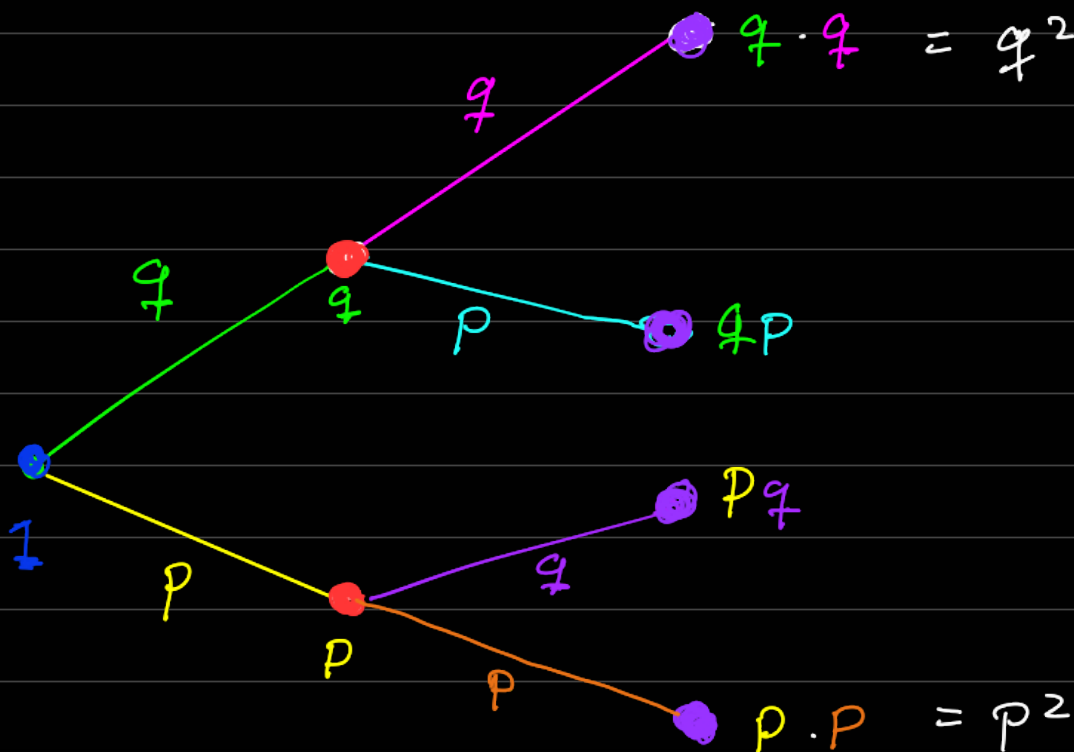
as we "follow"  $\rightarrow$  we see we have

$$\frac{1}{4} = \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) = p \cdot q$$

First step  
(Heads)

Second step  
(tails)

In general, we do not always have  $p = q = 1/2$   
so instead this tree looks like this for  
general  $p, q$ :  $(p+q) = 1$



\* Hopefully this makes sense to you and  
you can see how the probabilities of  
different outcomes multiply.

Also note that:

• The two results sum to 1 (as they should)

$$q + p = 1$$

• These Four sum to 1 as well:

$$\begin{aligned} q \cdot q + q p + p q + p p &= q^2 + 2qp + p^2 \\ &= (q + p)^2 = (1)^2 = 1 \end{aligned}$$

\* Now, this was all for us to justify that if we take  $n_R$  steps to the right and  $n_L$  to the left then the probability of this occurring once is simply:

$$\underbrace{p \dots p}_{n_R} \underbrace{q \dots q}_{n_L} = p^{n_R} q^{n_L}$$

example: Coin again: we said  $p = q = 1/2$  and said the probability of getting Heads  $\rightarrow$  Tails or HT is simply  
 $P(HT) = p \cdot q = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$  etc...

(ii) Now, I mentioned that we have a large number of ways to get to the same final spot for  $N = n_R + n_L$ . When we are looking at probabilities this means we are undercounting. Let's see why:

ex:  $N = 3$ : For 3 total steps we have  
 $3 = n_R + n_L$

We can have 8 different outcomes

$n_R \quad | \quad n_L \quad | \quad n$

|                                                                                                                                                                                                                                                          |   |   |    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|----|
|                                                                                                                                                                             | 3 | 0 | 3  |
| <br>                                                                                     | 2 | 1 | 1  |
| <br><br> | 1 | 2 | -1 |
|                                                                                                                                                                          | 0 | 3 | -3 |

↑  $1 + 3 + 3 + 1 = 8$

\* These distinct cases can be expressed mathematically via

$$\binom{N}{n_R} = \frac{N!}{n_R! (N - n_R)!}$$

but since  $N - n_R = n_L$  we have

$$\binom{N}{n_R} = \frac{N!}{n_R! n_L!}$$

said as "N choose  $n_R$ ".

Your notes have

$$\underline{C(a, b)} = \binom{a}{b} = \frac{a!}{b! (a - b)!}$$

\* Lets translate the above example. We saw for  $N = 3$  and some choice of  $n_R$  that there were different outcomes.

(a)  $N = 3, n_R = 3$  :  $\binom{3}{3} = C(3, 3) = \frac{3!}{3! 0!} = 1$

One possibility as we saw above

$$\textcircled{b} N=3, n_R=2: \binom{3}{2} = C(3,2) = \frac{3!}{2!1!} = \frac{3 \cdot 2}{2 \cdot 1} = 3$$

Three -different outcomes (as we saw)

etc....

\* We therefore have the <sup>total</sup> $\sqrt{N}$  probability of taking  $n_R$  steps to the right and  $n_L = N - n_R$  to the left is

$$P_N(n_R) = \left( \frac{N!}{n_R! n_L!} \right) \cdot p^{n_R} q^{n_L}$$

(Number of ways to take  $n_R$  to the right)

(Prob of taking  $n_R + n_L$  steps)

$P_N(n_R)$  is called the Binomial Distribution with

$$\sum_{n_R} P_N(n_R) = 1.$$

\* Cool, now let's relate the notation from above for those in your notes

### 5.1.1 Lottery

Recall:  $\binom{a}{b} = C(a,b)$  is the number of ways of choosing  $b$  elements out of a total of  $a$  elements.

\* Now, since order does not matter then  
... there are:  $C(49,6) = 49!$

$C(49, 6)$ : ways to choose 6 numbers out of 49.

• Since only one of these can be true the probability is simply

$$P = \frac{1}{C(49, 6)} \sim \frac{1}{14 \times 10^6}$$

\* Again, we know  $C(49, 6)$  is the total possibilities, now, the number of ways to choose 3 items from the 6, so that they match the 3 winning #'s is  $C(6, 3)$  and there are  $C(43, 3)$  ways to choose the other 3 numbers from 43 losing numbers.

Thus:

$$P = \frac{C(6, 3) C(43, 3)}{C(49, 6)}$$

$$C(43, 3) = \frac{43!}{3! 40!} \quad | \quad C(49, 6) = \frac{49!}{6! 43!}$$

$$\frac{C(43, 3)}{C(49, 6)} = \frac{\left( \frac{43!}{3! 40!} \right)}{\left( \frac{49!}{6! 43!} \right)} = \frac{43! 6! (43!)}{3! 40! (49!)}$$

Recall:  $43! = 43 \cdot 42 \cdot 41 \cdot 40 \dots 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$   
 $49! = 49 \cdot 48 \cdot 47 \dots 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$

$$\text{so } \frac{43!}{49!} = \frac{43 \cdot 42 \cdot 41 \dots 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{49 \cdot 48 \cdot 47 \dots 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}$$

$$= \frac{1}{\dots} \quad *$$

$$\frac{43!}{40!} = \frac{43 \cdot 42 \cdot 41}{1}$$

$$\text{and } \frac{6!}{3!} = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1} = \underline{6 \cdot 5 \cdot 4}$$

$$\frac{C(43, 3)}{C(49, 6)} = \frac{6 \cdot 5 \cdot 4 \cdot 43 \cdot 42 \cdot 41}{49 \cdot 48 \cdot 47 \cdot 46 \cdot 45 \cdot 44}$$

$$P = \frac{C(43, 3)}{C(49, 6)} \cdot C(6, 3) = \frac{6 \cdot 5 \cdot 4 \cdot 43 \cdot 42 \cdot 41}{49 \cdot 48 \cdot 47 \cdot 46 \cdot 45 \cdot 44} \cdot C(6, 3)$$

(as in the notes)

\* Note: why aren't we using the fancy expression from above for  $P$ ? Well, the above was for a binary system. Here we have a larger system i.e. 1 out of 40 instead of 1 out of two.

\* Use above to better understand Poker section and the meaning of  $C(N, m)$ .

→ Most texts use  $\binom{N}{m}$   
instead of  $C(N, m)$

